

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

1. Make a fast callable object of $\frac{x^3 - y + 2}{y^3 - x + 2}$ and draw the expression tree.
2. Let `s = lambda n: float(sum([exp(-(k/n)**2)*sin(2*k/n) for k in range(n)])/n)` be a function to approximate $\int_0^1 \exp(-x^2) \sin(2x) dx$.
 - (a) Time the execution of `s` for $n = 10000$. Explain why `s` is inefficient.
 - (b) Apply vectorization to improve the efficiency. Verify the correctness. Time the execution of the vectorized function for $n = 10000$, compare with timings of `s`.